CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Eigenvalues, Eigenvectors and Diagonalisation

Mark scheme

14 questions · 162 marks

EXAM BOARDS AQA 7367 · CIE 9231

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	Total	**162**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$(3-\lambda)(-2-\lambda)-2k=0$ $\lambda=4 \Rightarrow 6-2k=0$	**M1**	Forms $\det(\mathbf{A}-\lambda\mathbf{I})=0$ with $\lambda=4$ substituted at any stage, or uses $\mathbf{Av}=4\mathbf{v}$, to reach an equation in k. Condone sign slips.
	$k=3$	**A1**	
	$\lambda^2-\lambda-12=0 \Rightarrow \lambda=\ldots$	**M1**	Solves the characteristic equation with their k by any means, calculator allowed, or uses sum of eigenvalues = trace.
	$\lambda=-3$	**A1**	
	$\lambda=4$: $\begin{pmatrix}-1 & \text{“3”}\\ 2 & -6\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\mathbf{0} \Rightarrow x=3y$ $\lambda=-3$: $\begin{pmatrix}6 & \text{“3”}\\ 2 & 1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\mathbf{0} \Rightarrow y=-2x$	**M1**	Forms $(\mathbf{A}-\lambda\mathbf{I})\mathbf{v}=\mathbf{0}$ or $\mathbf{Av}=\lambda\mathbf{v}$ with their k and one of their eigenvalues, reaching a relation between x and y.
	$\begin{pmatrix}3\\1\end{pmatrix}$ for $\lambda=4$, $\begin{pmatrix}1\\-2\end{pmatrix}$ for $\lambda=-3$	**A1**	Any non-zero multiples, each matched to its eigenvalue.
	Total for Question 1	**6**	

Question	Scheme	Marks	Notes
2(a)	$\lambda = 1,\ \lambda = -1$	**B1**	Both values, and no others.
		[1]	
2(b)	$(a-\lambda)(b-\lambda) - 0.6^2 = 0$	**M1**	Forms $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$ in a, b, λ. Condone sign slips; need not be expanded.
	$\lambda^2 - (a+b)\lambda + ab - 0.36 \equiv \lambda^2 - 1$ $\Rightarrow a + b = 0,\ ab = -0.64 \Rightarrow a = \ldots$	**dM1**	Uses their eigenvalues, by comparing coefficients or substituting $\lambda = \pm 1$, to form two equations in a, b, and solves them simultaneously.
	$a = 0.8,\ b = -0.8$	**A1**	Both, oe. A0 if $a = -0.8$, $b = 0.8$ is not rejected.
		[3]	
2(c)	$\begin{pmatrix} \text{“}-0.2\text{”} & 0.6 \\ 0.6 & \text{“}-1.8\text{”} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\Rightarrow -0.2x + 0.6y = 0$	**M1**	Forms $(\mathbf{M} - \mathbf{I})\mathbf{v} = \mathbf{0}$ or $\mathbf{Mv} = \mathbf{v}$ with their numerical a, b, reaching a relation between x and y; or takes the perpendicular to an eigenvector for -1.
	$y = \frac{1}{3}x$	**A1**	Any equivalent equation, such as $x = 3y$. A direction vector alone scores A0.
		[2]	
	Total for Question 2	**6**	

Question	Scheme	Marks	Notes
3(a)	$(2-\lambda)(5-\lambda)-k^2=0$, $\lambda=1$: $4-k^2=0 \Rightarrow k=\ldots$	**M1**	Substitutes $\lambda=1$ into $\det(\mathbf{M}-\lambda\mathbf{I})=0$ and reaches an equation in k. Condone sign slips.
	$k=-2$	**A1**	$k=-2$ only; A0 for $k=\pm 2$ without selecting.
	$\lambda^2-7\lambda+6=0 \Rightarrow \lambda=\ldots$	**M1**	Obtains the other eigenvalue using their k: solves their characteristic quadratic by any means, or uses the trace or determinant.
	$\lambda=6$	**A1**	
	$(\mathbf{M}-\lambda\mathbf{I})\begin{pmatrix}x\\y\end{pmatrix}=\mathbf{0}$ $\lambda=1$: $x-2y=0$; $\lambda=6$: $2x+y=0$	**M1**	Forms $(\mathbf{M}-\lambda\mathbf{I})\mathbf{v}=\mathbf{0}$ or $\mathbf{Mv}=\lambda\mathbf{v}$ with their k and either eigenvalue, reaching a relation between x and y.
	$\begin{pmatrix}2\\1\end{pmatrix}$, $\begin{pmatrix}1\\-2\end{pmatrix}$	**A1**	Both correct, each any non-zero multiple, matched to its eigenvalue.
		[6]	
3(b)(i)	"$\begin{pmatrix}2\\1\end{pmatrix}$" $\cdot$ "$\begin{pmatrix}1\\-2\end{pmatrix}$" $=\ldots$	**M1**	Forms the scalar product of their two eigenvectors, or the product of the gradients of the two invariant lines.
	$=2-2=0$, so the invariant lines are perpendicular	**A1**	Scalar product 0, or gradient product -1, with the conclusion that the lines are perpendicular.
		[2]	
3(b)(ii)	One-way stretch	**B1**	"Stretch" alone accepted.
	scale factor 6, parallel to $y=-2x$	**B1**	Direction may be given as $\begin{pmatrix}1\\-2\end{pmatrix}$ or as perpendicular to $y=\frac{1}{2}x$.
	line of invariant points $y=\frac{1}{2}x$	**B1**	Any equivalent equation, such as $x=2y$.
		[3]	
	Total for Question 3	**11**	

Question	Scheme	Marks	Notes
4(a)	$\mathbf{P} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$, $\mathbf{D} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	**B1**	Eigenvectors in any order as columns, or nonzero multiples of them, with eigenvalues on the diagonal of $\mathbf{D}$ in the same order.
	$\mathbf{P}^{-1} = \frac{1}{\det \mathbf{P}} \operatorname{adj} \mathbf{P} = \ldots$ $\left[= \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & -1 \end{pmatrix} \right]$	**M1**	Finds $\mathbf{P}^{-1}$ for their $\mathbf{P}$ by cofactors, row reduction or calculator. Condone one slip in a cofactor or sign.
	$\mathbf{A} = \mathbf{PD}$"$\mathbf{P}^{-1}$" $= \begin{pmatrix} 2 & 0 & 1 \\ 0 & -1 & 1 \\ 2 & -1 & 1 \end{pmatrix}$ "$\mathbf{P}^{-1}$" $= \ldots$	**M1**	Multiplies their $\mathbf{P}$, $\mathbf{D}$ and $\mathbf{P}^{-1}$ in that order to a 3×3 matrix. Condone arithmetic slips.
	$\mathbf{A} = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -2 \\ 2 & -1 & 0 \end{pmatrix}$	**A1**	cao. A correct $\mathbf{A}$ without $\mathbf{P}^{-1}$ and the product $\mathbf{PDP}^{-1}$ seen scores B1 only.
		[4]	
4(b)	$\mathbf{A}^4 = \mathbf{PD}^4$"$\mathbf{P}^{-1}$" $= \begin{pmatrix} 16 & 0 & 1 \\ 0 & 1 & 1 \\ 16 & 1 & 1 \end{pmatrix}$ "$\mathbf{P}^{-1}$" $= \ldots$	**M1**	Uses $\mathbf{A}^4 = \mathbf{PD}^4\mathbf{P}^{-1}$ with their matrices, or multiplies their $\mathbf{A}$ from (a), including repeated squaring. Condone arithmetic slips.
	$\mathbf{A}^4 = \begin{pmatrix} 1 & -15 & 15 \\ 0 & 1 & 0 \\ 0 & -15 & 16 \end{pmatrix}$	**A1**	cao.
		[2]	
	Total for Question 4	**6**	

Question	Scheme	Marks	Notes
5(a)(i)	$\mathbf{M}\mathbf{v}_1 = \begin{pmatrix} 2 \\ a+10 \\ 2b-8 \end{pmatrix} = \lambda_1 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ $\Rightarrow \lambda_1 = 2$	**B1**	
		[1]	
5(a)(ii)	$a+10 = 2 \times$ "2", $\quad 2b-8=0$ $\Rightarrow a = \ldots,\ b = \ldots$	**M1**	Equates at least one of the second and third components of $\mathbf{M}\mathbf{v}_1$ with that of their $\lambda_1\mathbf{v}_1$ and solves. Condone a slip in the product.
	$a=-6, \quad b=4$	**A1**	
		[2]	
5(b)	$\mathbf{M}\mathbf{v}_2 = \begin{pmatrix} 4 & -1 & 1 \\ -6 & 5 & -6 \\ -8 & 4 & -5 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ $= \ldots$	**M1**	Multiplies $\mathbf{M}$, with their a and b, by $\mathbf{v}_2$; or uses the trace or determinant with their other eigenvalues. Condone one slip.
	$\mathbf{M}\mathbf{v}_2 = \begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix} \Rightarrow \lambda_2 = -1$	**A1**	
	$\begin{pmatrix} 1 & -1 & 1 \\ -6 & 2 & -6 \\ -8 & 4 & -8 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0}$ $\Rightarrow x-y+z=0,\ -6x+2y-6z=0$ $\Rightarrow x:y:z = \ldots$	**M1**	Forms $(\mathbf{M}-3\mathbf{I})\mathbf{v}=\mathbf{0}$ or $\mathbf{M}\mathbf{v}=3\mathbf{v}$ with their a, b and solves two equations for the component ratios, or takes a vector product of two rows. Condone one slip.
	$\mathbf{v}_3 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$	**A1**	Any nonzero multiple.
		[4]	
5(c)	$\mathbf{U} = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$ $\mathbf{D} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$	**B1**	Columns of $\mathbf{U}$ nonzero multiples of the three correct eigenvectors in any order, with the eigenvalues in $\mathbf{D}$ in matching order.
	$\det \mathbf{U} = 1(-1) - 0 + 1(2)\ [=1]$ $\mathbf{U}^{-1} = \frac{1}{\det \mathbf{U}} \operatorname{adj} \mathbf{U} = \ldots$	**M1**	Attempts the inverse of their 3×3 $\mathbf{U}$, e.g. determinant with matrix of cofactors transposed, or row reduction. Condone one slip; calculator inverse accepted.

Question 5 continued

Question	Scheme	Marks	Notes
	$\mathbf{U}^{-1} = \begin{pmatrix} -1 & 1 & -1 \\ 2 & -1 & 2 \\ 2 & -1 & 1 \end{pmatrix}$	**A1**	Correct inverse of their correct **U**, whatever its ordering or scaling.
		[3]	
	Total for Question 5	**10**	

Question	Scheme	Marks	Notes
6(a)	$\det(\mathbf{M}+2\mathbf{I}) = \begin{vmatrix} 3 & -2 & 2 \\ -2 & 4 & 0 \\ 2 & 0 & 2 \end{vmatrix}$ $= 3(8)+2(-4)+2(-8)$	**M1**	Evaluates $\det(\mathbf{M}+2\mathbf{I})$ by any expansion, or their $\det(\mathbf{M}-\lambda\mathbf{I})$ at $\lambda=-2$, or shows $\mathbf{Mv}=-2\mathbf{v}$ for a nonzero $\mathbf{v}$.
	$=0$, so -2 is an eigenvalue	**A1**	cso, with a conclusion. A0 if the conclusion is missing.
	$-2x+4y=0,\ 2x+2z=0 \Rightarrow \ldots$	**M1**	Forms at least two equations from $(\mathbf{M}+2\mathbf{I})\mathbf{v}=\mathbf{0}$ or $\mathbf{Mv}=-2\mathbf{v}$ and solves for a nonzero vector.
	$\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$	**A1**	Any nonzero multiple.
		[4]	
6(b)	$\det(\mathbf{M}-\lambda\mathbf{I}) = (1-\lambda)(\lambda^2-2\lambda)$ $+4\lambda-4(2-\lambda)$	**M1**	Expands $\det(\mathbf{M}-\lambda\mathbf{I})$ by any row or column to a cubic in λ. Condone sign slips.
	$\lambda^3-3\lambda^2-6\lambda+8=0$	**A1**	oe, such as its negative, $=0$ may be implied.
	$(\lambda+2)(\lambda^2-5\lambda+4)=0 \Rightarrow \lambda=\ldots$	**M1**	Obtains a quadratic factor of their cubic using $\lambda+2$ and solves it. Correct roots from a correct cubic imply the mark.
	$\lambda=1,\ 4$	**A1**	
	$\lambda=1$: $-2y+2z=0,\ -2x+y=0 \Rightarrow \ldots$	**M1**	Forms at least two equations from $(\mathbf{M}-\lambda\mathbf{I})\mathbf{v}=\mathbf{0}$ with their eigenvalue in place of 1 and solves for a nonzero vector.
	$\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$	**A1**	Any nonzero multiple.
	$\lambda=4$: $-2x-2y=0,\ 2x-4z=0 \Rightarrow \ldots$	**M1**	Forms at least two equations from $(\mathbf{M}-\lambda\mathbf{I})\mathbf{v}=\mathbf{0}$ with their eigenvalue in place of 4 and solves for a nonzero vector.
	$\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$	**A1**	Any nonzero multiple.
	Divides each by its length: $\sqrt{4+1+4}=\sqrt{1+4+4}=\sqrt{4+4+1}=3$	**M1**	Divides each of their three eigenvectors by its magnitude $\sqrt{x^2+y^2+z^2}$.

Question 6 continued

Question	Scheme	Marks	Notes
	$\mathbf{U} = \frac{1}{3}\begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & -2 \\ -2 & 2 & 1 \end{pmatrix}$	**A1**	Columns in any order, any column negated; decimals awrt 3 s.f. accepted.
	$\mathbf{D} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$	**B1**	Diagonal entries -2, 1, 4 ordered to match the eigenvectors in their **U**; zeros elsewhere.
		[11]	
	Total for Question 6	**15**	

Question	Scheme	Marks	Notes
7(a)	$\left(\frac{3}{8}-\lambda\right)^2-\frac{5}{16}\times\frac{5}{4}=0$ $\Rightarrow 4\lambda^2-3\lambda-1=0\Rightarrow\lambda=\ldots$	**M1**	Forms $\det(\mathbf{A}-\lambda\mathbf{I})=0$ as a quadratic and solves it by any means, calculator allowed. Condone sign slips in forming it.
	$\lambda=1,\ \lambda=-\frac{1}{4}$	**A1**	
	$\lambda=1$: $-\frac{5}{8}x+\frac{5}{16}y=0\Rightarrow y=\ldots$	**M1**	Uses $(\mathbf{A}-\lambda\mathbf{I})\mathbf{v}=\mathbf{0}$ or $\mathbf{Av}=\lambda\mathbf{v}$ with one of their eigenvalues to obtain a relation between x and y.
	$\begin{pmatrix}1\\2\end{pmatrix}$	**A1**	Any nonzero multiple, paired with $\lambda=1$.
	$\lambda=-\frac{1}{4}$: $\frac{5}{8}x+\frac{5}{16}y=0\Rightarrow\begin{pmatrix}1\\-2\end{pmatrix}$	**A1**	Any nonzero multiple, paired with $\lambda=-\frac{1}{4}$.
		[5]	
7(b)	$\mathbf{P}=\begin{pmatrix}1&1\\2&-2\end{pmatrix},\ \mathbf{D}=\begin{pmatrix}1&0\\0&-\frac{1}{4}\end{pmatrix}$	**B1ft**	ft their eigenvalues and eigenvectors, with the columns of $\mathbf{P}$ in the same order as the eigenvalues in $\mathbf{D}$.
		[1]	
7(c)	$\mathbf{A}^n=\mathbf{PD}^n\mathbf{P}^{-1}$, $\mathbf{D}^n=\begin{pmatrix}1&0\\0&\left(-\frac{1}{4}\right)^n\end{pmatrix}$	**M1**	Uses $\mathbf{A}^n=\mathbf{PD}^n\mathbf{P}^{-1}$ with their $\mathbf{P}$ and $\mathbf{D}$, raising the diagonal entries of $\mathbf{D}$ to the power n.
	$\mathbf{P}^{-1}=-\frac{1}{4}\begin{pmatrix}-2&-1\\-2&1\end{pmatrix}$ $\mathbf{A}^n=$ "$\mathbf{P}$"$\mathbf{D}^n$"$\mathbf{P}^{-1}$"$=\ldots$	**dM1**	Finds their $\mathbf{P}^{-1}$ by any method and multiplies the three matrices in the correct order to a single matrix. Condone arithmetic slips.
	$\mathbf{A}^n=\begin{pmatrix}\frac{1}{2}+\frac{1}{2}\left(-\frac{1}{4}\right)^n&\frac{1}{4}-\frac{1}{4}\left(-\frac{1}{4}\right)^n\\1-\left(-\frac{1}{4}\right)^n&\frac{1}{2}+\frac{1}{2}\left(-\frac{1}{4}\right)^n\end{pmatrix}$	**A1**	oe, such as $(-1)^n4^{-n}$ for $\left(-\frac{1}{4}\right)^n$.
	$\left\|-\frac{1}{4}\right\|<1\Rightarrow\left(-\frac{1}{4}\right)^n\to 0$	**E1**	States $\left(-\frac{1}{4}\right)^n\to 0$ because its modulus is less than 1, or oe. Not for "n is large" alone.
	$\mathbf{L}=\begin{pmatrix}\frac{1}{2}&\frac{1}{4}\\1&\frac{1}{2}\end{pmatrix}$	**A1ft**	ft their $\mathbf{A}^n$ with each $\left(-\frac{1}{4}\right)^n$ term set to 0.
		[5]	
7(d)	"$\mathbf{L}$" $\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\frac{1}{2}x+\frac{1}{4}y\\x+\frac{1}{2}y\end{pmatrix}$	**M1**	Finds the image of a general point (x,y) under their $\mathbf{L}$, or uses the columns of their $\mathbf{L}$ as image directions.

Question 7 continued

Question	Scheme	Marks	Notes
	$y = 2x$	**A1**	oe, such as $2x - y = 0$.
		[2]	
	Total for Question 7	**13**	

Question	Scheme	Marks	Notes
8(a)	$\det(\mathbf{A}-\lambda\mathbf{I})$ $=(-4-\lambda)[(-7-\lambda)(16-\lambda)+132]$ $+13[-6(16-\lambda)+72]$ $+8[132-12(7+\lambda)]$	**M1**	Expands $\det(\mathbf{A}-\lambda\mathbf{I})$ along any row or column, each of three terms an element times its 2×2 minor, or by Sarrus' rule. Condone sign and arithmetic slips.
	$-\lambda^3+5\lambda^2-2\lambda-8=0$ $\Rightarrow \lambda^3-5\lambda^2+2\lambda+8=0$	**A1**	Printed equation from a correct expansion without error; collected steps accepted.
	$\lambda=-1$: $-1-5-2+8=0$ $(\lambda+1)(\lambda^2-6\lambda+8)=0\Rightarrow\lambda=\ldots$	**M1**	Shows a root by substitution and obtains a quadratic factor, by division or comparing coefficients, with an attempt to solve it. Condone coefficient slips.
	$\lambda=-1,\ 2,\ 4$	**A1**	
		[4]	
8(b)	$\lambda=-1$: $-3x-13y+8z=0$, $-6x-6y+6z=0$ $\Rightarrow x:y:z=\ldots$	**M1**	Forms $(\mathbf{A}-\lambda\mathbf{I})\mathbf{v}=\mathbf{0}$ for one of their eigenvalues and solves two equations to a ratio $x:y:z$, or takes a vector product of two rows. Condone arithmetic slips.
	$\lambda=-1$: $\begin{pmatrix}1\\1\\2\end{pmatrix}$	**A1**	Each A1: any nonzero multiple; eigenvectors in any order.
	$\lambda=2$: $\begin{pmatrix}1\\2\\4\end{pmatrix}$	**A1**	
	$\lambda=4$: $\begin{pmatrix}1\\0\\1\end{pmatrix}$	**A1**	
	$\mathbf{P}=\begin{pmatrix}1&1&1\\1&2&0\\2&4&1\end{pmatrix}$, $\mathbf{D}=\begin{pmatrix}-1&0&0\\0&2&0\\0&0&4\end{pmatrix}$	**B1ft**	ft their eigenvalues and eigenvectors: eigenvectors as the columns of $\mathbf{P}$, eigenvalues in the same order on the diagonal of $\mathbf{D}$.
		[5]	
8(c)	$\mathbf{A}^n=\mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$ $\mathbf{D}^n=\begin{pmatrix}(-1)^n&0&0\\0&2^n&0\\0&0&4^n\end{pmatrix}$	**M1**	Uses $\mathbf{A}^n=\mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$ with their diagonal $\mathbf{D}$ raised to the power n entrywise. No proof of the result needed.
	$\mathbf{P}^{-1}=\begin{pmatrix}2&3&-2\\-1&-1&1\\0&-2&1\end{pmatrix}$	**B1**	Correct inverse of their valid $\mathbf{P}$.

Question 8 continued

Question	Scheme	Marks	Notes
	$(-1)^n \times 2 + 2^n \times (-1) + 4^n \times 0$ $= 2(-1)^n - 2^n$	**A1 AG**	Correct substituted sum of three products from valid $\mathbf{P}$, $\mathbf{D}^n$, $\mathbf{P}^{-1}$, leading without error to $2(-1)^n - 2^n$; full matrix product accepted.
		[3]	
	Total for Question 8	**12**	

Question	Scheme	Marks	Notes
9(a)	$\det(\mathbf{M}-\lambda\mathbf{I})$ $=(3-\lambda)[(3-\lambda)(-2-\lambda)+4]$ $\quad -4[(-2-\lambda)+1]-4[4-(3-\lambda)]$	**M1**	Expands $\det(\mathbf{M}-\lambda\mathbf{I})$ or $\det(\lambda\mathbf{I}-\mathbf{M})$ along any row or column, with three terms each a 2×2 cofactor. Condone sign slips.
	$=(3-\lambda)(\lambda^2-\lambda-2)-4(-1-\lambda)-4(1+\lambda)$	**A1**	Correct expansion, need not be simplified.
	$-\lambda^3+4\lambda^2-\lambda-6=0$ $\Rightarrow \lambda^3-4\lambda^2+\lambda+6=0$	**A1 AG**	Correct expansion simplified without error to the printed equation, set equal to zero; combined steps accepted.
		[3]	
9(b)	$f(-1)=-1-4-1+6=0$ $\Rightarrow(\lambda+1)(\lambda^2-5\lambda+6)=0\Rightarrow\lambda=\ldots$	**M1**	Shows a root of the cubic and factorises to a linear factor times a quadratic, by inspection, division or comparing coefficients, then solves the quadratic.
	$\lambda=-1,\ 2,\ 3$	**A1**	
		[2]	
9(c)	$(\mathbf{M}-\lambda\mathbf{I})\mathbf{v}=\mathbf{0}$, e.g. $\lambda=-1$: $x+y-z=0,\ x+4y-z=0\Rightarrow x:y:z=\ldots$	**M1**	Forms $(\mathbf{M}-\lambda\mathbf{I})\mathbf{v}=\mathbf{0}$ or $\mathbf{Mv}=\lambda\mathbf{v}$ for one of their eigenvalues and solves for a ratio $x:y:z$, or takes the vector product of two rows. Condone arithmetic slips.
	$\lambda=-1$: $\begin{pmatrix}1\\0\\1\end{pmatrix}$	**A1**	Any one correct eigenvector, any nonzero multiple. The three A1s may be earned in any order.
	$\lambda=2$: $\begin{pmatrix}0\\1\\1\end{pmatrix}$	**A1**	
	$\lambda=3$: $\begin{pmatrix}1\\1\\1\end{pmatrix}$	**A1**	
		[4]	
9(d)	$\mathbf{P}=\begin{pmatrix}1&0&1\\0&1&1\\1&1&1\end{pmatrix}$, $\mathbf{D}=\begin{pmatrix}-1&0&0\\0&2&0\\0&0&3\end{pmatrix}$	**B1ft**	ft their eigenvalues and eigenvectors: eigenvectors as columns of $\mathbf{P}$, eigenvalues on the diagonal of $\mathbf{D}$ in the same order.

Question 9 continued

Question	Scheme	Marks	Notes
	$\mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$ $\mathbf{D}^n = \begin{pmatrix} (-1)^n & 0 & 0 \\ 0 & 2^n & 0 \\ 0 & 0 & 3^n \end{pmatrix}$, $\mathbf{P}^{-1} = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & -1 \end{pmatrix}$	**M1**	Uses $\mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$ with $\mathbf{D}^n$ the diagonal of powers of their eigenvalues and an attempt at the inverse of their $\mathbf{P}$. Condone one slip in $\mathbf{P}^{-1}$.
	$\mathbf{P}\mathbf{D}^n = \begin{pmatrix} (-1)^n & 0 & 3^n \\ 0 & 2^n & 3^n \\ (-1)^n & 2^n & 3^n \end{pmatrix}$ $\Rightarrow \mathbf{M}^n =$ "$\mathbf{P}\mathbf{D}^n$" "$\mathbf{P}^{-1}$" $= \ldots$	**dM1**	Multiplies their $\mathbf{P}$, $\mathbf{D}^n$ and $\mathbf{P}^{-1}$ in that order to a single 3×3 matrix. Condone arithmetic slips.
	$\mathbf{M}^n = \begin{pmatrix} 3^n & 3^n-(-1)^n & (-1)^n-3^n \\ 3^n-2^n & 3^n & 2^n-3^n \\ 3^n-2^n & 3^n-(-1)^n & (-1)^n+2^n-3^n \end{pmatrix}$	**A1**	oe, e.g. $(-1)^{n+1}$ for $-(-1)^n$.
		[4]	
Total for Question 9		**13**	

Question	Scheme	Marks	Notes
10(a)	$\mathbf{B}(\mathbf{M}^{-1}\mathbf{e}) = \mathbf{M}^{-1}\mathbf{A}\mathbf{M}\mathbf{M}^{-1}\mathbf{e}$	**M1**	Replaces $\mathbf{B}$ by $\mathbf{M}^{-1}\mathbf{A}\mathbf{M}$ in $\mathbf{B}\mathbf{M}^{-1}\mathbf{e}$, or premultiplies $\mathbf{A}\mathbf{e} = \lambda\mathbf{e}$ by $\mathbf{M}^{-1}$ and inserts $\mathbf{M}\mathbf{M}^{-1}$. Matrix order must be correct.
	$= \mathbf{M}^{-1}\mathbf{A}\mathbf{e} = \mathbf{M}^{-1}(\lambda\mathbf{e})$	**A1**	Uses both $\mathbf{M}\mathbf{M}^{-1} = \mathbf{I}$ and $\mathbf{A}\mathbf{e} = \lambda\mathbf{e}$; combined steps accepted.
	$= \lambda(\mathbf{M}^{-1}\mathbf{e})$ $\mathbf{e} \neq \mathbf{0}$, $\mathbf{M}$ non-singular $\Rightarrow \mathbf{M}^{-1}\mathbf{e} \neq \mathbf{0}$	**A1 AG**	$\lambda(\mathbf{M}^{-1}\mathbf{e})$ reached without error, and $\mathbf{M}^{-1}\mathbf{e} \neq \mathbf{0}$ stated with a reason. Nothing for working backwards from the result.
		[3]	
10(b)	$\lambda = 3,\ -2,\ 1$	**B1**	
	$\lambda = 3$: $x - 5y = 0,\ 2x + 5y - 2z = 0 \Rightarrow \mathbf{v} = \ldots$	**M1**	Forms $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ or $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ with one of their eigenvalues and attempts a non-zero solution. Condone arithmetic slips.
	$\lambda = 3$: $\begin{pmatrix}10\\2\\15\end{pmatrix}$; $\lambda = -2$: $\begin{pmatrix}0\\3\\-5\end{pmatrix}$	**A1**	Any two correct, each paired with its eigenvalue; any non-zero multiples.
	$\lambda = 1$: $\begin{pmatrix}0\\0\\1\end{pmatrix}$	**A1**	All three correct, each paired with its eigenvalue; any non-zero multiples.
		[4]	
10(c)	$\det\mathbf{M} = 1$, cofactors $\begin{pmatrix}1&0&0\\-1&1&-2\\0&0&1\end{pmatrix}$	**M1**	Finds $\det\mathbf{M}$ and attempts the cofactor matrix, transposing and dividing; condone sign slips. Any valid method, e.g. row reduction. Implied by correct $\mathbf{M}^{-1}$.
	$\mathbf{M}^{-1} = \begin{pmatrix}1&-1&0\\0&1&0\\0&-2&1\end{pmatrix}$	**A1**	
	Eigenvalues of $\mathbf{B}$: $3,\ -2,\ 1$	**B1**	Not if found from $\mathbf{B}$ explicitly.
	"$\mathbf{M}^{-1}$" "$\begin{pmatrix}10\\2\\15\end{pmatrix}$" $= \ldots$, etc.	**M1**	Multiplies their 3×3 $\mathbf{M}^{-1}$ by at least one of their non-zero eigenvectors of $\mathbf{A}$. Not for eigenvectors found from $\mathbf{B}$ explicitly.

Question 10 continued

Question	Scheme	Marks	Notes
	3: $\begin{pmatrix} 8 \\ 2 \\ 11 \end{pmatrix}$; -2: $\begin{pmatrix} -3 \\ 3 \\ -11 \end{pmatrix}$; 1: $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$	**A1**	All three correct, each paired with its eigenvalue; any non-zero multiples.
		[5]	
	Total for Question 10	**12**	

Question	Scheme	Marks	Notes
11(a)	$\mathbf{A}^2\mathbf{e} = \mathbf{A}(\lambda\mathbf{e})$ $\mathbf{A}^{-1}\mathbf{A}\mathbf{e} = \lambda\mathbf{A}^{-1}\mathbf{e}$	**M1**	Premultiplies $\mathbf{A}\mathbf{e} = \lambda\mathbf{e}$ by $\mathbf{A}$ or $\mathbf{A}^{-1}$ to start either proof. Not for stating the results alone.
	$\mathbf{A}^2\mathbf{e} = \lambda\mathbf{A}\mathbf{e} = \lambda^2\mathbf{e}$	**A1**	One complete correct proof, for $\mathbf{A}^2$ or $\mathbf{A}^{-1}$, identifying eigenvalue and eigenvector; for the inverse, justify $\lambda \neq 0$ before dividing.
	$\mathbf{A}$ non-singular, $\mathbf{e} \neq \mathbf{0} \Rightarrow \lambda \neq 0$ $\mathbf{e} = \lambda\mathbf{A}^{-1}\mathbf{e} \Rightarrow \mathbf{A}^{-1}\mathbf{e} = \lambda^{-1}\mathbf{e}$	**A1 AG**	Both proofs complete without error, identifying eigenvalues and eigenvector. Needs $\lambda \neq 0$ justified from $\mathbf{A}$ non-singular and $\mathbf{e} \neq \mathbf{0}$ before dividing.
		[3]	
11(b)	$(\mathbf{A}^2 + 2\mathbf{A}^{-1})\mathbf{e} = \lambda^2\mathbf{e} + 2\lambda^{-1}\mathbf{e}$ $= (\lambda^2 + 2\lambda^{-1})\mathbf{e}$	**B1**	Applies the matrix to $\mathbf{e}$, uses both results of (a) and factorises, with a conclusion.
		[1]	
11(c)	$\begin{pmatrix} 3 & 1 & -1 \\ 3 & 3 & -3 \\ 3 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{0}$ $3x + y - z = 0,\ x + y - z = 0 \Rightarrow \ldots$	**M1**	Forms equations from $(\mathbf{A} + \mathbf{I})\mathbf{v} = \mathbf{0}$ or $\mathbf{A}\mathbf{v} = -\mathbf{v}$ and eliminates to relate the components, or crosses two rows of $\mathbf{A} + \mathbf{I}$. Condone one arithmetic slip.
	$\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$	**A1**	Any non-zero multiple.
		[2]	
11(d)	$\mathbf{A}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $\mathbf{A}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$	**M1**	Multiplies $\mathbf{A}$ by each given vector and compares with a multiple of it. Condone one arithmetic slip.
	Eigenvalues 1 and 2	**A1**	Each matched to its vector.
		[2]	
11(e)	$\mu = \lambda^2 + 2\lambda^{-1}$, $\lambda = -1, 1, 2 \Rightarrow \mu = \ldots$	**M1**	Substitutes at least two of their eigenvalues of $\mathbf{A}$ from (c) and (d) into $\lambda^2 + 2\lambda^{-1}$.
	$\mu = -1,\ 3,\ 5$	**A1**	
	"-1"^{2}, "3"^{2}, "5"^{2} $[= 1,\ 9,\ 25]$	**M1**	Squares their eigenvalues of $\mathbf{A}^2 + 2\mathbf{A}^{-1}$, or evaluates $(\lambda^2 + 2\lambda^{-1})^2$, for the diagonal of $\mathbf{D}$.

Question 11 continued

Question	Scheme	Marks	Notes
	$\mathbf{P} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 25 \end{pmatrix}$	**A1**	cso. Columns of $\mathbf{P}$ any non-zero multiples of the three eigenvectors, with $1, 9, 25$ in $\mathbf{D}$ in matching order.
		[4]	
	Total for Question 11	**12**	

Question	Scheme	Marks	Notes
12(a)(i)	$\det(\mathbf{M}-\lambda\mathbf{I}) = \frac{1}{27}\big[(2-3\lambda)\big((2-3\lambda)^2+2\big)$ $+\big(2(2-3\lambda)-1\big)+2\big(4+(2-3\lambda)\big)\big]$	**M1**	Expands the determinant of $\mathbf{M}-\lambda\mathbf{I}$ or $\lambda\mathbf{I}-\mathbf{M}$ along any row or column to a cubic in λ. Condone sign slips and a lost factor $\frac{1}{27}$.
	$\lambda^3-2\lambda^2+2\lambda-1=0$	**A1**	Must have leading coefficient 1 as printed; $-\lambda^3+2\lambda^2-2\lambda+1=0$ alone scores A0.
	$1-2+2-1=0 \Rightarrow 1$ is an eigenvalue	**A1 AG**	Substitutes $\lambda=1$ into the correct equation, shows it gives 0 and concludes 1 is an eigenvalue.
		[3]	
12(a)(ii)	$(\lambda-1)(\lambda^2-\lambda+1)=0 \Rightarrow \lambda=\ldots$	**M1**	Obtains a quadratic factor $\lambda^2+b\lambda+c$ of their cubic and solves it by formula, completing the square or calculator. Condone slips in the factor.
	$\lambda=\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$	**A1**	Both; exact or awrt $0.5\pm0.866i$.
	$b^2-4ac=-3<0$, so 1 is the only real eigenvalue	**E1**	States the discriminant of the correct quadratic is negative, or its roots are non-real, and concludes 1 is the only real eigenvalue.
		[3]	
12(b)	Diagonal entries of $\mathbf{D}$ are the eigenvalues of $\mathbf{M}$: $1,\ \frac{1}{2}\pm\frac{\sqrt{3}}{2}i$	**E1**	States the diagonal entries of $\mathbf{D}$ are the eigenvalues of $\mathbf{M}$, or that $\mathbf{M}$ and $\mathbf{D}$ share eigenvalues. No proof of this needed.
	Two are not real, so $\mathbf{D}$ cannot be real: the student is wrong	**E1**	States two eigenvalues are non-real, so $\mathbf{D}$ would have non-real entries, contradicting the claim.
		[2]	
12(c)(i)	$(\mathbf{M}-\mathbf{I})\mathbf{e}=\mathbf{0}$: $-x-y+2z=0,\ 2x-y-z=0$ $\Rightarrow x:y:z=\ldots$	**M1**	Forms $(\mathbf{M}-\mathbf{I})\mathbf{e}=\mathbf{0}$ or $\mathbf{M}\mathbf{e}=\mathbf{e}$ and solves two equations for the ratio $x:y:z$, or takes a vector product of two rows of $\mathbf{M}-\mathbf{I}$. Condone slips.

Question 12 continued

Question	Scheme	Marks	Notes
	$\mathbf{e} = \begin{pmatrix}1\\1\\1\end{pmatrix}$	**A1**	Any non-zero multiple.
		[2]	
12(c)(ii)	$\mathbf{Me} = \mathbf{e}$: every point on the line through O along $\mathbf{e}$ is fixed, so this line is the axis	**E1**	States that eigenvalue 1 means each point on the line along $\mathbf{e}$ is fixed, not merely the line invariant, so that line is the axis.
	$\mathbf{r} = t\begin{pmatrix}1\\1\\1\end{pmatrix}$	**B1ft**	ft their $\mathbf{e}$ from (c)(i). Needs $\mathbf{r} =$; any parameter.
		[2]	
12(c)(iii)	An eigenvector maps to a parallel vector; an acute rotation turns every vector off the axis to a non-parallel direction	**E1**	States an eigenvector maps to a parallel vector, and that a rotation through an acute angle changes the direction of every vector not along the axis. An algebraic argument alone scores E0.
		[1]	
	Total for Question 12	**13**	

Question	Scheme	Marks	Notes
13(a)	$\det(\mathbf{A}-\lambda\mathbf{I}) = (6-\lambda)[(3-\lambda)(6-\lambda)-4]$ $+2[-2(6-\lambda)-2]+[-4-(3-\lambda)]$	**M1**	Expands $\det(\mathbf{A}-\lambda\mathbf{I})$ along any row or column as three terms with 2×2 determinants. Condone sign slips.
	$\lambda^3-15\lambda^2+63\lambda-49=0$	**A1**	oe, such as $-\lambda^3+15\lambda^2-63\lambda+49=0$.
	$\lambda=1:\ 1-15+63-49=0$, so 1 is an eigenvalue	**B1**	Substituted terms shown equal to 0, with a conclusion.
	$\lambda^3-15\lambda^2+63\lambda-49=(\lambda-1)(\lambda^2-14\lambda+49)$	**M1**	Divides their cubic by $\lambda-1$ by any method, reaching a quadratic factor.
	$=(\lambda-1)(\lambda-7)^2$ Repeated eigenvalue 7	**A1**	cso. Repeated root shown by $(\lambda-7)^2$ or a zero discriminant, and 7 stated.
		[5]	
13(b)	$5x-2y+z=0,\ -2x+2y+2z=0\Rightarrow\ldots$	**M1**	Forms $(\mathbf{A}-\mathbf{I})\mathbf{v}=\mathbf{0}$ and solves two equations for a non-zero vector, or takes the cross product of two rows. Condone sign slips.
	$\begin{pmatrix}1\\2\\-1\end{pmatrix}$	**A1**	Any non-zero multiple.
	$\mathbf{A}-7\mathbf{I}$: $-x-2y+z=0$ (each row) $\Rightarrow x+2y-z=0$	**M1**	Forms $\mathbf{A}-7\mathbf{I}$ and reduces its rows to a single equation in x, y, z. Condone a sign slip.
	e.g. $\begin{pmatrix}1\\0\\1\end{pmatrix}$	**A1**	Any non-zero vector satisfying $x+2y-z=0$.
	$p+2q-r=0,\ p+r=0\Rightarrow\ldots$	**M1**	Solves their single equation from $\mathbf{A}-7\mathbf{I}$ with a zero scalar product with their first eigenvector for 7; or crosses that eigenvector with the eigenvector for 1.
	$\begin{pmatrix}1\\-1\\-1\end{pmatrix}$	**A1**	Any non-zero vector satisfying $x+2y-z=0$ and perpendicular to their correct first eigenvector for 7.
		[6]	
13(c)	$\frac{1}{\sqrt{6}}\begin{pmatrix}1\\2\\-1\end{pmatrix}$, $\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\1\end{pmatrix}$, $\frac{1}{\sqrt{3}}\begin{pmatrix}1\\-1\\-1\end{pmatrix}$	**M1**	Divides each of their three eigenvectors by its magnitude. Condone one magnitude slip.

Question 13 continued

Question	Scheme	Marks	Notes
	$\mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \end{pmatrix}$	**A1**	Columns are mutually perpendicular unit eigenvectors, in any order and with any signs.
	$\mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{pmatrix}$	**B1ft**	Eigenvalues on the diagonal in the order of the columns; ft their $\mathbf{P}$.
		[3]	
Total for Question 13		**14**	

Question	Scheme	Marks	Notes
14(a)(i)	$\lambda = 3$	**B1**	
		[1]	
14(a)(ii)	$-2 + 4 - p = 3 \times 2$	**M1**	Equates the second component of $\mathbf{A}$ times the eigenvector to 2 times their eigenvalue. Condone a slip in forming the component.
	$p = -4$	**A1 AG**	Correct equation in p leading without error to $p = -4$.
		[2]	
14(a)(iii)	$\det(\mathbf{A} - \lambda\mathbf{I}) = (-3 - \lambda)[(2 - \lambda)(3 - \lambda) - 20]$ $- 2[-2(3 - \lambda) + 40]$ $- 2[10 - 10(2 - \lambda)]$	**M1**	Expands $\det(\mathbf{A} - \lambda\mathbf{I})$ with $p = -4$ along any row or column, or by any valid method. Condone sign and arithmetic slips.
	$\lambda^3 - 2\lambda^2 - 5\lambda + 6 = 0$	**A1**	oe, such as the negated cubic.
	$(\lambda - 3)(\lambda^2 + \lambda - 2) = 0 \Rightarrow \lambda = \ldots$	**M1**	Solves their cubic by any means, such as dividing by $\lambda - 3$ and solving the quadratic; a calculator is allowed.
	$\lambda = 1,\ \lambda = -2$	**A1**	
	$\lambda = 1$: $-4\alpha + 2\beta - 2\gamma = 0$ $-2\alpha + \beta - 4\gamma = 0 \Rightarrow \ldots$	**M1**	Forms $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ for one of their other eigenvalues and solves at least two equations, or takes a cross product of two rows.
	$\lambda = 1$: $(1,\ 2,\ 0)^{\mathrm{T}}$	**A1**	Or any non-zero multiple.
	$\lambda = -2$: $(0,\ 1,\ 1)^{\mathrm{T}}$	**A1**	Or any non-zero multiple.
		[7]	
14(b)	$\mathbf{P}$ with columns $(1, 2, -1)^{\mathrm{T}}, (1, 2, 0)^{\mathrm{T}}, (0, 1, 1)^{\mathrm{T}}$ $\mathbf{D} = \mathrm{diag}(3,\ 1,\ -2)$	**B1ft**	ft their eigenvectors and eigenvalues: columns of $\mathbf{P}$ in the same order as the eigenvalues on the diagonal of $\mathbf{D}$. Any order.
		[1]	
14(c)(i)	$\int \frac{1}{u}\,\mathrm{d}u = \int k\,\mathrm{d}t \Rightarrow \ln\lvert u\rvert = kt + c$	**M1**	Separates the variables and integrates both sides, reaching $\ln u$ or $\ln\lvert u\rvert$.

Question 14 continued

Question	Scheme	Marks	Notes
	$u = Ke^{kt}$	**A1**	$u = Ke^{kt}$, K any real constant, including zero; oe. $u = e^{kt+c}$ alone is not the general solution.
		[2]	
14(c)(ii)	Uses $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$: $\dot{u} = 3u,\ \dot{v} = v,\ \dot{w} = -2w$	**M1**	Uses $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$ to reach three separate equations of the form $\dot{u} = \lambda u$ with their eigenvalues.
	$u = C_1e^{3t},\ v = C_2e^{t},\ w = C_3e^{-2t}$	**A1**	Three distinct arbitrary constants; rates in an order consistent with their $\mathbf{D}$.
	$x = u + v,\ y = 2u + 2v + w,\ z = -u + w$	**dM1**	Multiplies their $\mathbf{P}$ by their exponential u, v, w to obtain x, y, z. Not for using $\mathbf{P}^{-1}$.
	$x = C_1e^{3t} + C_2e^{t}$ $y = 2C_1e^{3t} + 2C_2e^{t} + C_3e^{-2t}$ $z = -C_1e^{3t} + C_3e^{-2t}$	**A1**	oe, including vector form; consistent with their order of eigenvectors.
		[4]	
14(c)(iii)	$C_1 + C_2 = 3,\ 2C_1 + 2C_2 + C_3 = 7$ $-C_1 + C_3 = -1 \Rightarrow C_1 = \ldots$	**M1**	Sets $t = 0$ in their general solution with three constants, equates to $(3, 7, -1)$ and solves; a calculator is allowed.
	$x = 2e^{3t} + e^{t}$ $y = 4e^{3t} + 2e^{t} + e^{-2t}$ $z = -2e^{3t} + e^{-2t}$	**A1**	
		[2]	
	Total for Question 14	**19**	